Group B: Machine Learning (410242)

Unit-wise viva questions with short spoken-style answers, then questions on practicals 7–11.

Unit I: Introduction to Machine Learning

Must know: ML vs traditional programming, AI vs ML vs DS, four types of learning, geometric/probabilistic/logical models, grouping vs grading, parametric vs non-parametric, data formats, learnability; case study: increasing Uber's sales.

Q1. What is machine learning? The field that gives computers the ability to learn from data without being explicitly programmed. Tom Mitchell's definition: a program learns from experience E for task T and performance measure P if its performance on T, measured by P, improves with E.

Q2. ML vs traditional programming? Traditional: data + rules (program) → output. ML: data + expected outputs → rules (the model). ML suits problems where the rules are too complex to write by hand, like spam detection or face recognition.

Q3. AI vs ML vs Data Science? AI is the broad goal of machines acting intelligently. ML is a subset of AI that learns patterns from data. Deep learning is a subset of ML using deep neural networks. Data science extracts insight from data using statistics, ML and visualisation, and overlaps all of them.

Q4. Supervised learning? Learning from labelled data (input → known output). Classification predicts a category (spam/not spam); regression predicts a number (fare). Algorithms: linear regression, KNN, SVM, decision trees, neural networks.

Q5. Unsupervised learning? Finding structure in unlabelled data: clustering (K-Means, customer segmentation), association rules (market basket), dimensionality reduction (PCA).

Q6. Semi-supervised learning? Uses a small amount of labelled data plus a large amount of unlabelled data, useful when labelling is expensive (medical images, web pages). Techniques: self-training, label propagation.

Q7. Reinforcement learning? An agent learns by interacting with an environment and receiving rewards or penalties, aiming to maximise cumulative reward. Key terms: agent, environment, state, action, reward, policy. Examples: game playing (AlphaGo), robotics, Q-learning.

Q8. Geometric models? Models that use geometry of the instance space: distances, lines and planes. Examples: linear regression (a line), SVM (a hyperplane), KNN (distances).

Q9. Probabilistic models? Models that represent uncertainty with probability distributions and predict P(Y|X). Examples: Naïve Bayes, Bayesian networks, logistic regression, Gaussian mixture models.

Q10. Logical models? Models expressed as if–then rules that partition the instance space. Examples: decision trees, rule learners. They are easy to interpret.

Q11. Grouping vs grading models? Grouping models split the instance space into segments and predict the same output for every instance in a segment (decision trees). Grading models assign each instance its own continuous score with no fixed segments (linear models, SVM).

Q12. Parametric vs non-parametric models? Parametric models have a fixed number of parameters regardless of data size (linear regression, logistic regression, Naïve Bayes); they are fast but assume a form. Non-parametric models grow in complexity with the data (KNN, decision trees, SVM with RBF); they are flexible but need more data.

Q13. Common data formats in ML? Structured or tabular (CSV, SQL), semi-structured (JSON, XML), and unstructured (text, images, audio, video). Features can be numerical (continuous or discrete) or categorical (nominal or ordinal).

Q14. What is learnability (PAC learning)? Whether a concept can be learned from a reasonable number of examples. PAC (Probably Approximately Correct) learning: with high probability $(1 - \delta)$, the learned hypothesis has error at most ε, using a sample size that is polynomial in $1/\varepsilon$ and $1/\delta$.

Q15. Statistical learning approaches? Frequentist (estimate parameters from data, e.g. maximum likelihood) and Bayesian (combine prior beliefs with data to get a posterior, e.g. MAP). Also generative (model P(X,Y)) vs discriminative (model P(Y|X)) approaches.

Q16. Case study: how would you increase Uber's sales with ML? First understand the requirement and collect data (trips, fares, time, location, weather, ratings). Then: predict demand by area and time to position drivers, use dynamic (surge) pricing from fare prediction (Practical 7), segment customers for targeted offers, and predict churn to retain riders.

Unit II: Feature Engineering

Must know: feature, normalization vs standardization, handling missing values, dimensionality reduction, PCA, Kernel PCA, LBP, feature selection (SFS, SBS), statistical features, MDS, matrix factorization.

Q1. What is a feature? An individual measurable property used as model input, i.e. a column of the dataset (age, glucose, distance). The output column is the target or label. Feature engineering means creating, transforming and selecting features to improve the model.

Q2. Why preprocess data? Real data has missing values, noise, outliers, inconsistent scales and text categories. Most algorithms need clean numeric input, and distance- or gradient-based algorithms need scaled features.

Q3. Normalization (min-max scaling)? $x' = (x - \min)/(\max - \min)$, which maps values to $[0, 1]$. It is sensitive to outliers. Used for neural networks and image pixels.

Q4. Standardization (z-score)? $z = (x - \mu)/\sigma$, giving mean 0 and standard deviation 1. It handles outliers better than min-max and is used for KNN, SVM, PCA and logistic regression (StandardScaler in sklearn).

Q5. Why fit the scaler only on training data? To avoid data leakage: test data must stay unseen. Fit on train, then transform both train and test with the same parameters.

Q6. How do you handle missing values? Drop rows or columns (if few or useless); impute with the mean (numeric, no outliers), the median (numeric with outliers) or the mode (categorical); forward/backward fill for time series; or predict them using KNN imputation or regression.

Q7. How do you handle categorical features? Label encoding for ordinal categories (low < medium < high); one-hot encoding for nominal ones (Geography → Germany/Spain columns), dropping one column to avoid the dummy-variable trap.

Q8. What is the curse of dimensionality? As the number of features grows, data becomes sparse, distances lose meaning, the data needed grows exponentially, and models overfit. Dimensionality reduction is the remedy.

Q9. Explain PCA. Principal Component Analysis is an unsupervised linear transformation. Standardize the data, compute the covariance matrix, find its eigenvectors and eigenvalues, sort by eigenvalue, and project the data onto the top k eigenvectors (principal components). The components are orthogonal and capture the maximum variance.

Q10. How do you choose the number of principal components? Keep enough components to explain about 95% of the variance (cumulative explained variance ratio), or use a scree plot (look for its elbow).

Q11. What is Kernel PCA? PCA in a higher-dimensional feature space reached through a kernel function (RBF, polynomial), without computing that space explicitly (the kernel trick). It captures non-linear structure that normal PCA misses.

Q12. What is Local Binary Pattern (LBP)? A texture feature extractor for images. For each pixel, compare it with its 8 neighbours (1 if neighbour ⩾ centre, else 0) to get an 8-bit code; the histogram of codes over the image is the feature vector. Used in face recognition.

Q13. Feature selection vs feature extraction? Selection keeps a subset of the original features (interpretable). Extraction creates new features from combinations of the originals (PCA, LBP).

Q14. Types of feature selection techniques? Filter methods use statistical scores independent of the model (correlation, chi-square, ANOVA, mutual information). Wrapper methods test subsets with the model itself (SFS, SBS, RFE). Embedded methods select during training (Lasso, tree feature importance).

Q15. Sequential Forward Selection vs Sequential Backward Selection? SFS starts with no features and repeatedly adds the one that most improves model performance. SBS starts with all features and repeatedly removes the one whose removal hurts performance least. Both are greedy wrapper methods.

Q16. What are statistical (count/length/mean/median/mode) features? New features computed from raw data: counts (number of words in an email, purchases per customer), lengths (text length), mean/median/mode/standard deviation of values over a window or group. Practical 8′s word-count columns are count-based features.

Q17. What is Multidimensional Scaling (MDS)? A technique that places points in a low-dimensional space (usually 2-D) so that the pairwise distances between them are preserved as well as possible. Used for visualising similarity between items.

Q18. What are matrix factorization techniques? Decomposing a matrix into a product of smaller matrices: SVD ($A = U\Sigma V^T$, the basis of PCA and LSA), NMF (non-negative factors), and the user–item factorization used in recommender systems (Netflix ratings = user factors × movie factors).

Q19. Case study: how would you find which marketing campaign works? Use EDA (exploratory data analysis) and correlation of campaigns with sales; hypothesis testing (A/B tests, t-tests, chi-square) for statistical significance; feature importance from tree

models, Lasso or RFE to select key features; then segment customers to target them.

Unit III: Supervised Learning: Regression

Must know: bias, variance, generalization, underfitting, overfitting, linear regression, Lasso, Ridge, gradient descent, MAE, RMSE, R2; case study: stock price prediction.

Metric	Formula	Meaning
MAE	$(1/n)\ \Sigma\ \lvert y - \hat{y}\rvert$	Average absolute error, in the target's units; robust to outliers
MSE	$(1/n)\ \Sigma\ (y - \hat{y})^2$	Squares errors, so it punishes big errors
RMSE	$\sqrt{\text{MSE}}$	Same units as the target; most commonly reported
R2	$1 - SS_res / SS_tot$	Fraction of variance explained; 1 = perfect, 0 = no better than predicting the mean

Q1. What is regression? Supervised learning that predicts a continuous numeric value (price, temperature, fare) from input features.

Q2. What is bias? Error from overly simple assumptions: the model can't capture the true pattern. High bias leads to underfitting (e.g. a straight line fitted to curved data).

Q3. What is variance? Error from the model being too sensitive to the particular training data; small changes in data change the model a lot. High variance leads to overfitting.

Q4. Explain the bias–variance trade-off. Total error = bias² + variance + irreducible noise. Making a model more complex lowers bias but raises variance. The goal is the sweet spot with the lowest test error.

Q5. Underfitting vs overfitting? Underfitting: poor on both training and test data (model too simple). Overfitting: excellent on training data, poor on test data (the model memorised noise).

Q6. How do you reduce overfitting? Get more data, use a simpler model, add regularization (L1/L2), use cross-validation, select fewer features, stop early, use dropout (neural networks), and prune or limit depth (trees).

Q7. What is generalization? The model's ability to perform well on new, unseen data, not just on training data. It is measured on a held-out test set or with cross-validation.

Q8. Explain linear regression. It fits $y = b_0 + b_1x_1 + \ldots + b_nx_n$ by minimising the sum of squared errors (Ordinary Least Squares). Closed form: $\beta = (X^TX)^{-1}X^Ty$. Simple regression has one feature; multiple regression has many.

Q9. Assumptions of linear regression? A linear relationship, independent errors, homoscedasticity (constant error variance), normally distributed errors, and no multicollinearity among features.

Q10. What is Ridge regression? Linear regression plus an L2 penalty: minimise SSE + $\lambda\Sigma\beta^2$. It shrinks coefficients toward zero (never exactly zero), reduces variance and handles multicollinearity.

Q11. What is Lasso regression? Linear regression plus an L1 penalty: minimise SSE + $\lambda\Sigma|\beta|$. It can shrink coefficients exactly to zero, so it also performs feature selection. Elastic Net combines L1 and L2.

Q12. What does λ (alpha) control? The strength of regularization. $\lambda = 0$ gives plain linear regression; a large λ gives a very simple model (more bias, less variance). It is chosen with cross-validation.

Q13. What is gradient descent? An iterative optimisation algorithm: start with random weights, compute the gradient of the cost function, and update $w = w - \alpha \cdot \partial J/\partial w$, repeating until convergence. α is the learning rate.

Q14. What happens if the learning rate is too high or too low? Too high: steps overshoot the minimum and may diverge. Too low: convergence is very slow and can stall.

Q15. Batch vs stochastic vs mini-batch gradient descent? Batch uses all data per update (stable, slow). Stochastic (SGD) uses one sample per update (fast, noisy). Mini-batch uses small batches such as 32 samples (the usual compromise in deep learning).

Q16. MAE vs RMSE: when to prefer which? RMSE when large errors are especially bad (it squares them). MAE when you want robustness to outliers and an easily explained average error.

Q17. Can R2 be negative? Yes, on test data, when the model is worse than simply predicting the mean. Adjusted R2 penalises adding useless features.

Q18. Case study: stock market prediction? It is a time series regression: features are lagged prices, moving averages, volume and news sentiment; models range from linear regression and Random Forest to LSTM. Split by time (no shuffling) and evaluate with RMSE/MAE. Markets are noisy, so accuracy is limited.

Unit IV: Supervised Learning: Classification

Must know: KNN, SVM, bagging, boosting, Random Forest, AdaBoost, binary vs multiclass, imbalanced data, OvO vs OvA, accuracy/precision/recall/F-score, cross-validation, micro vs macro averaging; case study: thyroid disorders.

	Predicted Positive	Predicted Negative
Actual Positive	TP	FN (Type II error)
Actual Negative	FP (Type I error)	TN

Accuracy = (TP+TN)/total · Precision = TP/(TP+FP) · Recall = TP/(TP+FN) · F1 = 2PR/(P+R) · Specificity = TN/(TN+FP) · Error rate = 1 − accuracy.

Q1. What is classification? Supervised learning that predicts a discrete class label: binary (spam/not spam) or multiclass (thyroid type).

Q2. How does KNN work? Store all training data. For a new point, compute its distance to every training point (usually Euclidean), take the k nearest, and assign the majority class (or the average, for regression).

Q3. Why is KNN called lazy and non-parametric? Lazy: it doesn't train; all the work happens at prediction time. Non-parametric: it assumes no fixed form for the data, and the "model" is the whole dataset.

Q4. How do you choose k? Try several values and pick the best using cross-validation or a test-error plot. Small k: noisy, overfits. Large k: too smooth, underfits. Use odd k for binary problems to avoid ties; $\sqrt{n}$ is a common start.

Q5. Pros and cons of KNN? Simple, no training, handles multiclass naturally. But prediction is slow (O(n·d) per query), it needs feature scaling, it's sensitive to irrelevant features and outliers, and it suffers from the curse of dimensionality.

Q6. Distance measures used in KNN? Euclidean $\sqrt{\Sigma(x-y)^2}$, Manhattan $\Sigma|x-y|$, Minkowski (general p), Hamming for categorical or binary data, cosine for text.

Q7. What is SVM? Support Vector Machine finds the hyperplane $w \cdot x + b = 0$ that separates classes with the maximum margin. The training points closest to the hyperplane are the support vectors, and only they define it.

Q8. Hard margin vs soft margin, and what is C? Hard margin allows no misclassification (only works for perfectly separable data). Soft margin allows some violations through slack variables; C controls the penalty: large C gives a narrow margin and few errors (risk of overfitting), small C gives a wide margin (more regularized).

Q9. What is the kernel trick? Computing dot products in a high-dimensional feature space via a kernel function without ever transforming the data, so a linear separator there becomes a non-linear boundary in the original space. Kernels: linear, polynomial, RBF (Gaussian), sigmoid.

Q10. What is ensemble learning? Combining several models (weak learners) to get better accuracy and stability than any single model.

Q11. What is bagging? Bootstrap aggregating: train many models in parallel on random samples drawn with replacement, then combine them by voting (classification) or averaging (regression). It reduces variance.

Q12. How is Random Forest different from plain bagging? It bags decision trees and also picks a random subset of features at each split, which decorrelates the trees. It gives feature importance and out-of-bag error, and it rarely overfits badly.

Q13. What is boosting? Train models sequentially, each focusing on the errors of the previous ones, then combine them with weights. It reduces bias. Examples: AdaBoost, Gradient Boosting, XGBoost.

Q14. How does AdaBoost work? Start with equal sample weights; train a weak learner (usually a decision stump); increase the weights of misclassified samples; give each learner a vote weight $\alpha = \frac{1}{2} \ln((1-\text{err})/\text{err})$; the final prediction is the weighted vote of all learners.

Q15. Bagging vs boosting? Bagging: parallel, independent models, reduces variance (Random Forest). Boosting: sequential, dependent models, reduces bias, can overfit noisy data (AdaBoost).

Q16. One-vs-All vs One-vs-One? One-vs-All (One-vs-Rest) trains K binary classifiers, one per class against all others, and picks the most confident. One-vs-One trains $K(K-1)/2$ classifiers, one per pair of classes, and takes a majority vote (sklearn's SVC uses OvO).

Q17. What is an imbalanced dataset, and how do you handle it? One class is much rarer (fraud, disease, churn at 20%). Accuracy then misleads. Fixes: precision/recall/F1/ROC-AUC as metrics, oversampling (SMOTE), undersampling, class weights, stratified splits.

Q18. Precision vs recall: which matters more? Precision when false positives are costly (spam filter, so real mail isn't lost). Recall when false negatives are costly (cancer or diabetes screening, so no patient is missed).

Q19. What is k-fold cross-validation? Split the data into k parts; train on k−1 parts and test on the remaining one, rotating k times; average the scores. It gives a more reliable estimate than one split. Stratified k-fold keeps class ratios in each fold.

Q20. Micro vs macro averaging? Macro: compute the metric per class, then take the simple average (every class counts equally, so rare classes matter). Micro: pool all TP/FP/FN across classes, then compute (dominated by large classes; micro-F1 equals accuracy in single-label multiclass).

Q21. Case study: thyroid disorder prediction? It is a multiclass problem with 4 classes (hyperthyroid, hypothyroid, euthyroid-sick, euthyroid) from features like TSH, T3, T4, age and sex. The data is imbalanced (most patients are normal), so use SVM/Random Forest with OvO or OvA, class weights, and macro-averaged F1 with stratified cross-validation.

Unit V: Unsupervised Learning

Must know: K-Means, K-Medoids, hierarchical, density-based (DBSCAN) and spectral clustering; outliers with Isolation Forest and LOF; elbow method; intrinsic vs extrinsic evaluation; case studies: market basket analysis, customer segmentation.

Q1. What is clustering? Grouping unlabelled data so that points in the same cluster are similar (high intra-cluster similarity) and points in different clusters are dissimilar (low inter-cluster similarity).

Q2. Explain the K-Means algorithm. Choose k; initialise k centroids (k-means++ spreads them out); assign each point to its nearest centroid; recompute each centroid as the mean of its points; repeat until assignments stop changing. It minimises WCSS. Complexity $O(n \cdot k \cdot d \cdot \text{iterations})$.

Q3. Limitations of K-Means? You must choose k in advance; the result depends on initialisation (local minima); it assumes spherical, similar-sized clusters; it is sensitive to outliers and scaling; and it works only with numeric data.

Q4. K-Means vs K-Medoids? K-Medoids (PAM) uses actual data points (medoids) as centres and can use any distance measure. It's more robust to outliers than the mean-based K-Means, but slower ($O(k(n-k)^2)$ per iteration).

Q5. What is hierarchical clustering? Building a tree of clusters (a dendrogram). Agglomerative (bottom-up) starts with each point alone and merges the closest pair; divisive (top-down) starts with one cluster and splits it. No k is needed up front: you cut the dendrogram at a chosen height.

Q6. Linkage methods in hierarchical clustering? Single (minimum distance between clusters, gives chaining), complete (maximum distance, compact clusters), average (mean distance), Ward (merges the pair that least increases total variance; used in Practical 11).

Q7. What is DBSCAN? Density-based clustering with two parameters: eps (neighbourhood radius) and MinPts. Core points have ⩾ MinPts neighbours within eps; border points are near a core point; everything else is noise. It finds clusters of arbitrary shape, detects outliers, and needs no k.

Q8. What is spectral clustering? Build a similarity graph of the points, compute the eigenvectors of its graph Laplacian, and run K-Means on the points embedded in those eigenvectors. It handles non-convex shapes such as concentric circles.

Q9. What is an outlier, and why detect it? A data point that differs markedly from the rest (a measurement error, fraud, a rare event). Outliers distort means, regression lines and K-Means centroids. Detection is used for fraud, intrusion and fault detection.

Q10. How does Isolation Forest work? Build random trees that split on random features at random values. Outliers are isolated in fewer splits (short path lengths), so the anomaly score is based on average path length: shorter means more anomalous.

Q11. What is Local Outlier Factor (LOF)? It compares the local density of a point with the density of its k nearest neighbours. LOF ≈ 1 means normal; LOF ≫ 1 means much sparser than its neighbours, i.e. an outlier. It finds local outliers that global methods miss.

Q12. Explain the elbow method. Run K-Means for k = 1..10 and plot WCSS (inertia) against k. WCSS always falls as k grows; the “elbow” where the decrease suddenly flattens is the best k. In Practical 11 the drops were 4550, 1976, 831…, so k = 3.

Q13. Intrinsic vs extrinsic evaluation? Intrinsic (internal) measures use only the data and clusters: silhouette score, Davies–Bouldin index, Calinski–Harabasz index, WCSS. Extrinsic (external) measures compare clusters with true labels: purity, Rand index / Adjusted Rand Index, NMI, homogeneity, completeness.

Q14. What is the silhouette score? s = (b − a)/max(a, b), where a is the mean distance to points in the same cluster and b is the mean distance to the nearest other cluster. It ranges from −1 to 1; closer to 1 means well-separated clusters.

Q15. Customer segmentation? Cluster customers on features like age, income, spending score or RFM (recency, frequency, monetary value) using K-Means, then target each segment differently (premium offers, discounts, loyalty programs).

Q16. Market basket analysis? Finding items frequently bought together using association rules (Apriori, FP-Growth). Support = fraction of transactions containing the itemset; confidence = P(B|A); lift = confidence / P(B), where lift > 1 means a positive association. Example: bread → butter.

Unit VI: Introduction to Neural Networks

Must know: perceptron, MLP, backpropagation, FLANN, RBF network, activation functions, basics of RNN and CNN; case study: movie recommendation.

Activation	Formula	Range	Used in
Sigmoid	$1/(1+e^{-x})$	(0, 1)	Binary output layer
Tanh	$(e^{x}-e^{-x})/(e^{x}+e^{-x})$	(−1, 1)	Hidden layers, RNNs
ReLU	max(0, x)	[0, ∞)	Hidden layers (default)
Leaky ReLU	x if x > 0 else 0.01x	(−∞, ∞)	Fixes “dying ReLU”
Softmax	$e^{xi}/\Sigma e^{xj}$	(0, 1), sums to 1	Multiclass output layer
Step	1 if x ≥ 0 else 0	{0, 1}	Original perceptron

Q1. What is an artificial neural network? A model inspired by the brain, made of layers of connected neurons. Each neuron computes a weighted sum of its inputs plus a bias and passes it through an activation function: $y = f(\Sigma w_i x_i + b)$.

Q2. What is a single-layer perceptron? One layer of output neurons connected directly to the inputs, with a step activation. Learning rule: $w = w + \eta(\text{target} - \text{output})x$. It can learn only linearly separable functions (AND, OR).

Q3. Why can’t a single perceptron learn XOR? XOR isn’t linearly separable: no single straight line separates its classes. A hidden layer (an MLP) solves it.

Q4. What is a multilayer perceptron (MLP)? A feed-forward network with an input layer, one or more hidden layers and an output layer, using non-linear activations. With enough hidden neurons it can approximate any continuous function (the universal approximation theorem).

Q5. Explain backpropagation. Forward pass: compute outputs and the loss. Backward pass: use the chain rule to compute the gradient of the loss with respect to every weight, layer by layer from the output back to the input. Then update each weight by gradient descent, $w = w - \eta \cdot \partial L/\partial w$. Repeat for many epochs.

Q6. Why are activation functions needed? Without non-linear activations, any number of layers collapses into one linear transformation, so the network could only learn linear relationships.

Q7. What is the vanishing gradient problem? With sigmoid or tanh, gradients get multiplied by small derivatives ($\leqslant 0.25$ for sigmoid) through many layers and shrink toward zero, so early layers stop learning. Fixes: ReLU, good weight initialisation, batch normalization, residual connections, LSTM/GRU for sequences.

Q8. Epoch vs batch vs iteration? An epoch is one pass over the whole training set. A batch is the subset of samples used for one weight update. Iterations per epoch = samples / batch size.

Q9. What loss functions are used? MSE for regression; binary cross-entropy for two classes (Practical 9); categorical cross-entropy for multiclass with softmax.

Q10. What is a Functional Link ANN (FLANN)? A single-layer network that removes the hidden layer by expanding the input with non-linear functions (trigonometric sin/cos, polynomial or Chebyshev terms). It is simpler and faster to train than an MLP while still handling non-linear problems.

Q11. What is a Radial Basis Function (RBF) network? A three-layer network whose hidden neurons use Gaussian functions of the distance between the input and a centre, $\varphi = \exp(-\|x-c\|^2/2\sigma^2)$, with a linear output layer. Centres are often found by K-Means. It trains fast and is used for function approximation and classification.

Q12. What is a CNN? A Convolutional Neural Network for grid data such as images. Convolution layers slide small filters to detect local features (edges, then shapes, then objects); pooling layers reduce size; fully connected layers classify. Weight sharing means far fewer parameters than an MLP. Uses: image classification, face recognition, medical imaging.

Q13. What is an RNN? A Recurrent Neural Network for sequences (text, speech, time series). It keeps a hidden state that carries information from previous time steps: $h_t = f(Wx_t + Uh_{t-1})$. Plain RNNs suffer from vanishing gradients; LSTM and GRU use gates to remember long-term dependencies.

Q14. CNN vs RNN? CNNs capture spatial patterns in fixed-size inputs (images). RNNs capture temporal order in variable-length sequences (sentences, stock prices).

Q15. How do you prevent overfitting in neural networks? Dropout, early stopping, L2 regularization (weight decay), more data or data augmentation, batch normalization, and a smaller network. Practical 9 uses dropout and early stopping.

Q16. Case study: movie recommendation system? Collaborative filtering recommends what similar users liked (user–item rating matrix, matrix factorization, or a neural network with user and movie embeddings). Content-based filtering recommends movies similar to ones you liked (genre, cast). Hybrids combine both. Evaluate with RMSE on ratings, or precision@k.

Lab practicals 7–11: questions on your code

#	Practical	Algorithms	Your result
7	Uber fare prediction	Linear Regression, Random Forest	RF R2 0.742, RMSE $1.89; LR R2 0.673
8	Email spam	KNN, SVM (linear)	SVM 97.2% accuracy; KNN 87.2%
9	Bank churn	ANN (Keras)	About 86% accuracy (verify on your laptop)
10	Diabetes	KNN, k = 19	74.7% accuracy, precision 0.67, recall 0.56
11	Sales clustering	K-Means, hierarchical	Elbow at k = 3

Questions asked for every ML practical

- **Libraries used?** pandas (data loading), NumPy (arrays), matplotlib/seaborn (plots), scikit-learn (models, metrics, scaling), TensorFlow/Keras (Practical 9).
- **Why split train and test? What ratio?** To measure performance on unseen data; 80/20 with `random_state=42` for reproducibility.
- **What does `stratify=y` do?** Keeps the same class proportions in train and test.
- **`fit` vs `transform` vs `fit_transform`?** fit learns parameters (mean, std); transform applies them; fit_transform does both. Use fit_transform on train, transform on test.

- **What is `random_state`?** A seed so the random split or model gives the same result every run.

Practical 7: Uber fare

- **What preprocessing did you do?** Dropped `Unnamed: 0` and `key`, removed nulls, extracted hour/day/month/year/weekday from the datetime, and computed the trip distance with the Haversine formula.
- **What is the Haversine formula?** The great-circle distance between two latitude/longitude points on a sphere (Earth radius 6371 km).
- **How did you find outliers?** Box plots, then removed invalid values (fare ⩽ 0, passengers outside 1–6, impossible coordinates) and applied the IQR rule: outside Q1 − 1.5·IQR to Q3 + 1.5·IQR.
- **Which feature is most correlated with fare?** Distance (about 0.80).
- **Why does Random Forest beat Linear Regression?** It captures non-linear relationships and interactions, and averaging many trees reduces variance.
- **Why scale for Linear Regression but not Random Forest?** Trees split on thresholds, so feature scale doesn't matter to them.

Practical 8: Email spam

- **What do the 3000 columns mean?** The counts of the 3000 most common words in each email (bag of words).
- **Why did SVM do much better than KNN?** With 3000 dimensions, distances become less meaningful (curse of dimensionality), which hurts KNN; a linear SVM handles high-dimensional sparse text well.
- **Why a linear kernel?** Text data is usually close to linearly separable in high dimensions, and linear is fast.
- **What does C = 0.01 mean?** Strong regularization: a wider margin that tolerates some errors, which generalizes better here.
- **Which metric matters most for spam?** Precision: marking a real email as spam (a false positive) is worse than missing a spam.

Practical 9: Bank churn (neural network)

- **Which columns did you drop, and why?** RowNumber, CustomerId and Surname, because they are identifiers with no predictive value.

- **How did you handle Geography and Gender?** One-hot encoding with `drop_first=True`.
- **Why normalize for a neural network?** Features have very different ranges (Balance up to 250,000 vs 0/1 flags); scaling makes gradient descent converge faster and more stably.
- **Explain your network.** Input (11 features) → Dense 32 ReLU → Dropout 0.2 → Dense 16 ReLU → Dropout 0.2 → Dense 1 Sigmoid. Adam optimizer, binary cross-entropy loss.
- **What points of improvement did you implement?** More layers and neurons, dropout, early stopping (patience 10, restore best weights) and more epochs.
- **Why sigmoid in the output?** It gives a probability of churn between 0 and 1; the threshold is 0.5.
- **Is 86% accuracy good?** Only about 20% churn, so predicting "stays" for everyone already gives 80%. Recall on the churn class matters, and class weights or SMOTE could improve it.

Practical 10: KNN on diabetes

- **Why replace zeros?** Glucose, BloodPressure, SkinThickness, Insulin and BMI can't be 0; a 0 means missing, so it's replaced with the column median.
- **How did you choose k = 19?** 5-fold cross-validation over odd k from 1 to 25, picking the lowest error rate.
- **Why scale features for KNN?** It is distance-based; otherwise Insulin (0–846) dominates Pedigree (0–2.4).
- **Compute your metrics from the confusion matrix.** TN = 85, FP = 15, FN = 24, TP = 30: accuracy = 115/154 = 0.747, error rate = 0.253, precision = 30/45 = 0.667, recall = 30/54 = 0.556.
- **Why is recall important here?** A false negative means a diabetic patient is told they are healthy.

Practical 11: K-Means on sales data

- **Which features did you use, and why?** QUANTITYORDERED, PRICEEACH, SALES and MSRP: numeric buying behaviour. IDs, names, dates, phone numbers and addresses carry no behaviour.
- **Why `encoding="latin1"`?** The CSV contains non-UTF-8 characters, so the default reader throws a UnicodeDecodeError.

- **What is inertia / WCSS?** The sum of squared distances of points to their cluster centroid.
- **What does `n_init=10` mean?** K-Means runs 10 times with different starting centroids and keeps the best result.
- **What did your clusters mean?** They roughly matched small, medium and large deals (checked with a cross-tab against DEALSIZE).
- **What does the dendrogram show?** The order and distance at which clusters merge (Ward linkage); cutting at the largest vertical gap gives the number of clusters.

Mini Project: Titanic Survival Prediction

Your result: SVM (RBF) had the best 5-fold CV accuracy, 0.831, then Logistic Regression 0.827 and Random Forest 0.822. The tuned Random Forest reached CV 0.830. Women survived at 74% vs men at 19%, and 1st class at 63% vs 3rd class at 24%. The top features were Sex and Title_Mr.

- **Problem type?** Supervised binary classification: Survived 0 or 1.
- **What did EDA show?** “Women and children first” and a strong class effect: 1st-class passengers had better lifeboat access.
- **How did you handle missing Age (about 20%)?** Filled each passenger with the median age of their title group (a Master is a boy, about 4 years), which is better than one overall median.
- **Embarked, Fare and Cabin?** Embarked (2 missing) got the mode; Fare (1 missing in test) the median; Cabin (77% missing) became a HasCabin 0/1 flag, which indicates wealth.
- **What features did you engineer?** Title extracted from Name, FamilySize = SibSp + Parch + 1, IsAlone, HasCabin.
- **Why is Title so useful?** It encodes sex, age group and social status in one feature.
- **Why combine train and test before encoding?** So both get identical one-hot columns. Only the inputs are combined, never the labels, so there's no leakage.
- **Why isn't test.csv used for accuracy?** It has no Survived column (Kaggle keeps the answers). That's why 20% of train.csv is held out and cross-validation is used.
- **Why a pipeline with StandardScaler?** It scales inside each CV fold using only that fold's training data, which LR, KNN and SVM need.
- **What does GridSearchCV do?** It tries every hyperparameter combination with cross-validation and keeps the best (here max_depth = 4, 300 trees).

- **Why did the tuned model score lower on the 179-row validation split?** A single small split is noisy; tuning optimises the CV average, which is the fairer measure.
- **Read your confusion matrix.** TN = 96, FP = 14, FN = 22, TP = 47: recall for survivors is 0.68, lower than for deaths (0.87), because survivors are the minority (38%).
- **What is submission.csv?** Predictions (PassengerId, Survived) for the 418 test passengers, in the format Kaggle accepts.
- **How could you improve accuracy?** Gradient boosting (XGBoost), ticket-group survival features, fare per person, or a voting ensemble of the best models. About 80–83% is the realistic ceiling for honest models.